

Population-Scale Pattern Control Through Participation Mode Governance

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Attribution

This work derives from and formalizes the inter-Self coordination architecture introduced in “Inter-Self Coordination via Shared Substrate / Full Aspect Integration” (Li, April 2026), the third paper in the CKS theory series following “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026) and “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026).

Abstract

This note formalizes D2.62, an operational decomposition of Claim 6 (population-scale collective evolution) in the calibrated-humility register. It derives from D2.31 (participation governance dimensions) and D2.32 (pattern propagation mechanics). The note specifies four governance-configurable participation modes — Full Participation, Contribute-Only, Absorb-Only, and Restricted Participation — through which each Self controls its position in the population-scale pattern propagation network. It identifies the governance controls that actuate each mode, states the convergence-diversity control implication of aggregate participation mode distribution at population scope, identifies legitimate uses of each mode, and states the anti-pattern of participation mode gaming and the natural trust-calibration response to it. An operational test is provided: for a Self in the FAI network, whether an observer can determine its effective participation mode from its participation configuration and historical DNA absorption decisions.

1. D2.62 as Claim 6 Operational Decomposition

Claim 6 of Paper 3 specifies population-scale collective evolution as the architectural object produced when accumulated FAI events across many CKS-governed Selves operate under joint authority across population-level governance perimeters. The claim’s architectural commitment is that what propagates across the network, what diverges, what converges, and what is protected as specialization all remain substrate content under home-governance authority — no central coordinator determines population-level outcomes.

D2.62 addresses the operational face of that commitment: how each Self’s governance actually controls its participation in pattern propagation. D2.31 established the participation governance

dimensions — propagation permissions, propagation restrictions, temporal scope, partner scope — as configurable substrate content. D2.32 established how governance patterns propagate through governed absorption and re-contribution, event by event. D2.62 formalizes the four positions in the participation configuration space that those dimensions compose into, the governance controls that actuate each position, and the population-level implications of each position’s distribution across the network.

The calibrated-humility register applies throughout. Four modes are specified as governance-configurable positions available under the architecture’s design. Which mode any given Self operates in is a governance decision, not an architectural prescription. Population-scale dynamics follow from the aggregate of those decisions across the network; the architectural commitment is to what the inheritance produces compositionally, with specific population-level convergence or diversity trajectories awaiting empirical observation in actual multi-Self deployments.

2. Four Participation Modes

The four participation modes are positions in the two-dimensional space defined by contribution behavior (what the Self shares outward through FAI events) and absorption behavior (what the Self takes in from other Selves’ contributions through DNA evolution). They are not mutually exclusive over time — a Self may operate in different modes for different domains or event types — but at any given configuration, each participation dimension resolves to one of these positions.

Mode 1 — Full Participation

The Self both contributes governance patterns to the network through FAI aspect contribution and absorbs governance patterns from other Selves through DNA evolution absorption. Propagation permissions are set broadly; propagation restrictions are minimal; DNA absorption decisions at FAI events are consistently accepting within the home governance authority’s authorization. This is the default participation mode in the architecture’s design intention: the Self maximizes both its contribution to the population-scale collective evolution and its benefit from it. A Self in Mode 1 influences the network and is influenced by it across accumulated events.

Mode 2 — Contribute-Only

The Self contributes its governance patterns through FAI aspect contribution but does not absorb governance patterns from other Selves’ contributions. Propagation permissions are set broadly, enabling outward flow. DNA absorption is systematically declined — either through participation configuration that sets absorption scope to minimal or through consistent decline of DNA absorption opportunities at D2.11 — so the Self’s home governance content evolves independently of what other Selves carry. The Self influences the network without being influenced by it in this mode.

Mode 3 — Absorb-Only

The Self does not contribute its governance patterns — propagation permissions are set to narrow or minimal, so FAI events do not result in the Self's content flowing to others — but absorbs governance patterns from other Selves' contributions through DNA evolution absorption. The Self benefits from network learning without its own content entering the network's propagation flow. The Self's home substrate evolves under the influence of patterns accumulated elsewhere; the network does not acquire the Self's patterns in return.

Mode 4 — Restricted Participation

The Self participates in FAI events with specific partners for specific domains, but neither broadly contributes nor broadly absorbs. Propagation permissions and absorption scope are both set narrowly, bounded to particular partner configurations per D2.58's partner-specific governance. The Self's participation in pattern propagation is targeted rather than network-wide; its participation configuration enables coordination with selected Selves while limiting both outward and inward pattern flow to defined scopes.

3. Governance Controls That Actuate Each Mode

The four modes are not architecturally prescribed states with separate implementations. They are positions in the participation configuration space that each Self's governance produces through the interaction of three governance controls already specified in prior notes.

Propagation Permissions (D2.31 Dimension A) govern what the Self contributes to FAI events. Setting permissions broadly — authorizing sharing of DNA-layer content, schema patterns, and orchestration logic — enables the contribution behavior present in Mode 1 and Mode 2. Setting permissions narrowly or to minimal — restricting what aspects the Self contributes to FAI events or limiting sharing to a minimal agreed scope — suppresses outward contribution and enables the non-contribution behavior present in Modes 3 and 4.

Propagation Restrictions (D2.31 Dimension B) govern what the Self actively protects from outward flow. Where Dimension A sets the general permission scope, Dimension B specifies carve-outs within it — content treated as proprietary, content whose provenance must be carried if shared, content that cannot be shared regardless of event configuration. Comprehensive restrictions enable Mode 3 or Mode 4 by containing what would otherwise propagate under a broader permission setting.

DNA Absorption Governance (D2.11) governs what the Self absorbs from each FAI event's available contribution content. The absorption decisions made at each event — which contributed patterns from other Selves to incorporate into the home DNA layer — accumulate across events into the Self's historical absorption profile. Systematic decline of absorption at D2.11, across multiple events, implements the non-absorption behavior present in Mode 2 and Mode 4. Consistent acceptance within home-authorized scope implements the absorption behavior present in Modes 1 and 3.

The governance controls interact: a Self that sets permissions broadly but accepts systematically narrow absorption scope at D2.11 is in Mode 2. A Self that sets permissions narrowly but absorbs whenever other Selves contribute is in Mode 3. The mode is the joint product of the participation configuration and the pattern of absorption decisions, not of either control in isolation.

4. Convergence-Diversity Control at Population Scope

The aggregate distribution of participation modes across the population-scale network determines the network's convergence-diversity balance — the degree to which governance patterns homogenize across Selves over accumulated FAI events versus the degree to which divergence and domain-specific specialization are preserved.

A network in which most Selves operate in Mode 1 (Full Participation) exhibits stronger convergence pressure. Common governance patterns — schemas, orchestration logics, conflict-handling protocols that prove effective across many domains — propagate broadly because contribution is broad and absorption is broad. The cumulative effect of many Selves absorbing from each other across many events tends toward alignment of widely-applicable patterns while specializations that no other Self absorbs persist at home.

A network in which most Selves operate in Modes 2, 3, or 4 preserves more diversity. Mode 2 (Contribute-Only) Selves contribute without being influenced; their home content evolves independently even as it propagates. Mode 3 (Absorb-Only) Selves accumulate others' patterns without contributing their own, which means their home evolution may converge toward the network while not exporting their own specialization into it. Mode 4 (Restricted Participation) Selves limit both flows, preserving strong independence in both directions.

No single Self controls whether the population-scale network converges or diverges. The balance is a population-level property that emerges from the aggregate of governance choices across all participating Selves. The architecture enables governance choice of the balance — each Self's home governance can configure its participation mode — but does not prescribe the balance. This is the calibrated-humility register's honest expression of D1.27's convergence-with-diversity property at the operational level: the architectural commitment is to the governance mechanism that enables the choice; the empirical character of actual convergence-diversity trajectories in deployed multi-Self networks awaits observation.

5. Legitimate Uses of Each Mode

Each mode has governance justifications that make it an appropriate choice rather than a failure mode for certain Self configurations.

Mode 1 (Full Participation) is appropriate for a Self whose governance content is broadly applicable across domains and whose home governance has no specific restriction on sharing, and for whom benefits from others' accumulated governance approaches are likely to be relevant. It is

the default for Selves operating in similar domains where mutual learning is plausible.

Mode 2 (Contribute-Only) is appropriate for a Self that has developed governance approaches valuable to others but operates in a domain where other Selves' approaches are not relevant to its home context. A Self whose operational domain is sufficiently distinct that DNA absorption from other Selves would introduce governance patterns ill-suited to home conditions may legitimately configure for contribution without absorption. The home governance justification is domain specificity, not an intent to exploit the network.

Mode 3 (Absorb-Only) is appropriate for a Self in early stages of governance maturity — newly established, with little accumulated governance content worth contributing — that benefits from others' experience while its own contribution value is low. The home governance justification is developmental stage. A Self in this position may reasonably expect to transition toward Mode 1 as its governance content matures. Mode 3 is also appropriate where the Self's governance content is subject to confidentiality requirements that prevent contribution, while absorption remains authorized.

Mode 4 (Restricted Participation) is appropriate for a Self operating in a highly regulated domain where both broad contribution (which could expose regulated content) and broad absorption (which could introduce governance patterns from unvetted sources) are restricted by the governing organization's risk management requirements. Targeted participation with approved partners in approved domains is the governance-appropriate response to this constraint.

6. Anti-Pattern: Participation Mode Gaming

The legitimate uses stated in §5 share a common feature: each is grounded in a genuine governance justification that the home governance authority can articulate — domain specificity, developmental stage, regulatory constraint, risk management requirement. Each represents a configuration that the home governance chose for reasons related to its operational situation, not a configuration chosen to maximize benefit extraction from the network at minimal cost to it.

The anti-pattern is participation mode gaming: a Self that operates in Mode 3 (Absorb-Only) without a legitimate governance justification, configured to accumulate governance patterns propagated by other Selves while contributing nothing to the network in return. Over time, if many Selves adopt this configuration without justification, network quality degrades — the Selves whose Mode 1 or Mode 2 participation sustains the propagation network carry increasing burden while absorb-only Selves benefit from their contributions without reciprocation.

The architecture does not require central policing to discourage this pattern. Governance trust calibration (D2.29) naturally produces a self-correcting mechanism: a Self that never contributes builds no contribution history, and its governance trust score reflects this absence. Lower trust scores result in narrower sharing scope from other Selves — partners configure their propagation permissions to share less with low-trust Selves, or to share only under more restrictive

provenance-carry-over policies. The absorb-only gaming strategy thus generates diminishing returns: as the network's Selves calibrate trust and tighten sharing toward non-contributing Selves, the content available to those Selves through absorption narrows. The mechanism is incentive-structural rather than enforcement-structural; no governance authority needs to identify or sanction gaming Selves for the correction to operate.

7. Operational Test

For a Self in the FAI network, its effective participation mode is observable from the combination of its participation configuration (the propagation permissions and restrictions set in its substrate under D2.31) and its historical DNA absorption decisions (the pattern of acceptance and decline across accumulated FAI events under D2.11).

An observer with access to the Self's participation configuration substrate and its FAI event history can determine the Self's effective participation mode by answering two questions:

Question 1 — Contribution behavior: Does the Self's participation configuration set propagation permissions broadly (allowing substantive DNA-layer content, schema patterns, or orchestration logic to flow to FAI partners), or narrowly (restricting what the Self's contribution amounts to at FAI events)?

Question 2 — Absorption behavior: Across the Self's history of FAI events, does the Self's DNA evolution absorption reflect consistent acceptance of contributed patterns within home-authorized scope, or consistent decline?

If contribution is broad and absorption is accepting: Mode 1. If contribution is broad and absorption is consistently declined: Mode 2. If contribution is narrow and absorption is accepting: Mode 3. If contribution is narrow and absorption is declined or restricted to specific partners: Mode 4.

The test is behavioral and substrate-observable. It does not require access to the home governance authority's internal rationale for the configuration. It reads the configuration and history as substrate content — which is what the architecture requires them to be — and produces a mode determination from observable governance facts. Mixed configurations (different modes for different domains or different partner scopes) produce mixed readings; the test applies domain- or partner-scope-specifically where the configuration is differentiated.

A Self that cannot be mode-classified by this test is a Self whose participation configuration or absorption history is not accessible as substrate content, which is itself a governance finding warranting attention.

8. Conclusion

D2.62 formalizes the governance mechanism through which each Self controls its position in the population-scale pattern propagation network. Four participation modes — Full Participation,

Contribute-Only, Absorb-Only, and Restricted Participation — are governance-configurable positions, not architecturally prescribed states. The architecture supports all four; participation configuration and DNA absorption governance jointly determine which mode each Self operates in. Legitimate governance justifications exist for each mode. The convergence-diversity balance of the population-scale network is a property that emerges from the aggregate distribution of participation modes, not from any single Self's governance choice or from a central coordinator. Trust calibration provides a self-correcting anti-gaming mechanism without requiring enforcement governance. The operational test confirms that mode determination is substrate-observable.

Source Papers

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance*. April 2026. ORCID: 0009-0004-8065-3235.

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